

I Become Tentacle, We Become "Tentacus": A Mixed Reality Ritual for Distributed Bodies and Collective Minds

Yiyang Sun
China Academy of Art
Hangzhou, China
yiyangsun0610@gmail.com

Ziying Chen
China Academy of Art
Hangzhou, China
czy202304@163.com

La Luo
China Academy of Art
Hangzhou, China
ruoraaa@gmail.com

Botao Amber Hu*
Reality Design Lab
New York City, USA
botao@reality.design



Figure 1: *Tentacus* is a mixed reality ritual that explores decentralized intelligence and collective embodiment, where each dancer becomes a tentacle by wearing soft-textile gloves with embedded smartphones that function as both sensors and actuators. They negotiate movement through connection to a circular ring, becoming a collective, fluid tentacular being.

Abstract

Inspired by the octopus's distributed neurophysiology, *Tentacus* is a mixed reality artistic ritual that weaves multiple persons into a decentralized intelligence and collective embodiment. In this ritual, *Tentacus* "awakens"—a collective being composed of distributed bodies (dancers) that forms neither a centralized individual nor a mere aggregate, but rather a dynamic, fluid, tentacular intelligence. Each dancer becomes a tentacle: the left hand grips a circular ring, symbolizing continuous kinship; the right hand wears a soft-textile "tentacle" glove with an embedded smartphone whose LiDAR sensor functions as a spatial motion tracker and whose vibration motor acts as a vibrotactile actuator. Each right "tentacle" continuously detects environmental terrain through proximity sensing and broadcasts haptic waves to neighboring dancers' actuators. These proximity-touch data are transformed in real time into luminous point clouds, inscribed on a public display against a pre-scanned 3D Gaussian representation of the site, enabling spectators to witness *Tentacus* as it senses and remembers. Grounded in Donna Haraway's Chthulucene "tentacular thinking" and Margrit Shildrick's call to "Embody the Monster", the piece elevates felt knowing over visual dominance. It manifests a protocol of multi-person shared control

through distributed sensing, cross-body inference, and negotiated motion, nurturing more-than-human awareness.

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms**; Collaborative and social computing systems and tools.

Keywords

Decentralized Intelligence, Somaesthetic design, More-than-human Design, Playful Wearables, Octopus, Co-located Mixed Reality, Tentacular Thinking, Gaussian Splatting, Speculative Design

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1 Motivation

Octopus neurophysiology [Godfrey-Smith 2016] offers an alternative model of mind—decentralized intelligence. Nearly two-thirds of an octopus's neurons reside in its limbs rather than a centralized brain, unlike humans, whose perception–decision–action loop is centralized in our brains. Grounded in Donna Haraway's Chthulucene "tentacular thinking" [Haraway 2016], which challenges visually dominated, anthropomorphic thinking, we seek to create a multi-person ritual that embodies participants as tentacles with a shared-control protocol to develop synaesthetic "fingeryeyes" [Hayward 2010]—tentacular "visuality" sensations distributed across tentacles. Within the ritual, participants temporarily become collective

*Corresponding author; also serves as a visiting lecturer at the School of Design & Innovation, China Academy of Art.

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beings: *Tentacus*. This resonates with Margrit Shildrick's call to "Embody the Monster" [Shildrick 2002]: becoming "Tentacus" means accepting partial, depersonalized, distributed perception, where intention and motion are negotiated rather than owned. Through this somaesthetic ritual as participatory speculative design, participants rehearse more-than-human awareness [Poikolainen Rosén et al. 2025] and experiment with different ways of being.

2 Design and Experience

We propose a mixed reality ritual that embodies participants as a collective being through a multi-person shared-control protocol. Participants gather around a central circular ring representing an octopus's brain. With their left hands, they grip this ring, sensing collective movement intentions and negotiating motion with neighbors through subtle forces. Their right hands wear soft textile tentacle-shaped gloves with smartphones embedded under the palm (see Figure 2). These smartphones function as both spatial trackers and haptic devices, broadcasting over a local wireless network for low-latency communication and spatial co-location [Hu et al. 2023]. As participants scan the environment by moving their right hands near surfaces, the LiDAR scanner detects proximity to ground, walls, or objects and activates vibration signals proportional to that proximity. These signals transmit as haptic waves to the immediately left and right neighboring participants' smartphones and then propagate to further participants' smartphones, activating haptics through their motors, creating a relay system that enables cross-body perceptual synchronization. Through this relay, each participant feels their neighbors' environmental encounters while maintaining continuous physical connection over the ring with other participants using their left hands, establishing ongoing agency, intention and motion negotiation.



Figure 2: The glove mimics an octopus tentacle, crafted from soft textile with lightweight cotton lining for natural touch and flexibility. A smartphone is mounted beneath the palm, with its LiDAR scanner exposed to capture the surrounding environment.

Participant Journey. The ritual begins in stillness. Participants wear optional blindfolds to decenter vision and heighten tentacular feeling, embodying as "fingeryeyes." One tentacle brushes an environmental surface and sends subtle haptic signals to its neighbors; neighbors amplify these signals. Communication flows from

neighbor to neighbor. Intentions are constantly negotiated through forces on the ring—so no one leads. Over minutes, micro-intentions congeal into macro-behaviors: coiling, reaching, detouring. Participants feel both enlarged and partial—able to act only with others, yet more capable than alone. Agency emerges from interdependence—where "I become tentacle, we become Tentacus".

Spectator Journey. The site is pre-scanned using a smartphone with LiDAR and reconstructed using 3D Gaussian splatting technology [Fei et al. 2025]. During the performance, all smartphones on tentacles transmit 6-DoF spatial pose tracking and proximity signals to a centralized server that generates real-time glowing luminous point-cloud traces rendered on a public display. When tentacles brush terrain, contacts bloom as glowing points that decay over time. The public spectators can witness Tentacus sensing and remembering, externalized as a shared visual memory.

3 Conclusion

Tentacus offers a new "mixed reality" from the participant's perspective: their reality is defined not by vision but by a synaesthetic experience composed of haptics, forces, and ongoing negotiation between agency, intention, and motion. With its decentralized intelligence model based on distributed perception and mutual response, Tentacus breaks away from the traditional single "perception-decision-action" loop, instead forming a shared control logic of "perception aggregation and path coordination". It functions as a living protocol, exploring how collective minds form, enact, and entangle in somaesthetic ways that transcend individual agency.

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